# 7.2 Properties of Parallelograms

Name _____ Period ____ Date _____

Geometry 7.2 · Student sheet

**I can…**

- use the sides and angles of a parallelogram to find missing measures.
- use the fact that the diagonals of a parallelogram bisect each other.
- find a missing vertex or the intersection of the diagonals in the coordinate plane.

## Part 1 · Discovery: measure a parallelogram

Use the lines on graph paper to draw a parallelogram ABCD (two pairs of parallel sides). Draw both diagonals and label where they meet E. Measure.

| AB | BC | CD | DA |
|---|---|---|---|
| | | | |

| m∠A | m∠B | m∠C | m∠D |
|---|---|---|---|
| | | | |

| AE | EC | BE | ED |
|---|---|---|---|
| | | | |

Three things that are always true about a parallelogram:

## Part 2 · Notes: fill in the blanks

Opposite sides of a parallelogram are _____.

Opposite angles of a parallelogram are _____.

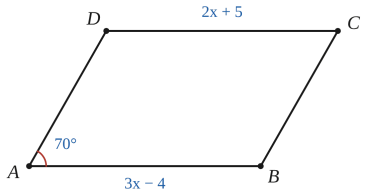
Consecutive angles of a parallelogram are _____ (they add to _____°).

The diagonals of a parallelogram _____ each other.

## Part 3 · Examples (work them with the class)

### Example 1

ABCD is a parallelogram. AB = 3x − 4, DC = 2x + 5, and m∠A = 70°. Find x, AB, m∠B, and m∠C.

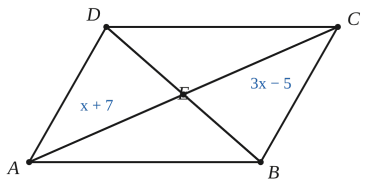



### Example 2

In parallelogram PQRS, m∠P = (4y + 10)° and m∠Q = (2y + 20)°. Find y and both angles.

### Example 3

In parallelogram ABCD, the diagonals meet at E. AE = x + 7 and EC = 3x − 5. Find x and AC.



### Example 4

Three vertices of parallelogram JKLM are J(−1, 2), K(3, 4), and L(5, 0). Find the coordinates of M and of the point where the diagonals meet.

## Part 4 · Partner practice

### Partner 1

In parallelogram QRST, the diagonals meet at E. QE = 2y + 1 and ES = 4y − 9. Find y and QS.

### Partner 2

In parallelogram PQRS, m∠P = (3x + 15)° and m∠R = (5x − 25)°. Find x, m∠P, and m∠Q.